

Accuracy: the assumption is that the data is accurate as it is DSAR-requested data of my actual trips.

Consistency: the city field contains 145 records (71%) that mention the country (e.g. “Birmingham, UK” (Figure 2). To keep the data consistent, I manually added the country names to all the other records (e.g. “Madrid, Spain”) (Figure 3).

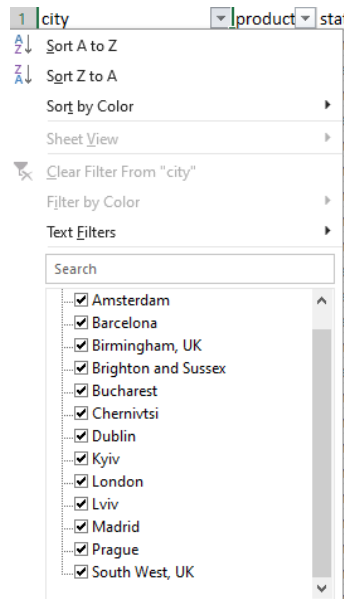


Figure 2: Inconsistency in Cities

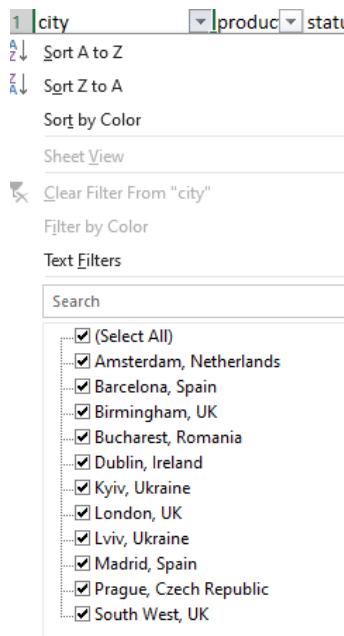


Figure 3: Cities and Countries

Validity: As shown in Figure 4, 32 records (16%) had issues with unintelligible entries and special characters in `begintrip_address` and/or `dropoff_address` for the trips that took place in Ukraine, Romania, and the Czech Republic. Since the Ukrainian language uses Cyrillic alphabet, and both Romanian and Czech contain special symbols and extra characters different from the English language, incorrect encoding might have caused the issue. Those records were removed from the dataset. An alternative approach could be to find the correct addresses using geographical coordinates in the `begintrip_lat` and `begintrip_lng` fields.

	A	B	C	D	E	F	G	H	I	J	K	L
1	city							begintrip_address	dr	d	dr	dropoff_address
12	Bucharest	ub rid	2024-	#	26.1					45	26	Calea BucureEutilor 224E, Otopeni 075150, Romania
13	Bucharest	ub rid	2024-	#	26.1					45	26	Calea BucureEutilor 224E, Otopeni 075150, Romania
14	Bucharest	ub con	20 20	#	26.1			Calea GriviiИei 143, BucureИ™ti, Romania	2024-	45	26	Calea BucureEuti 1K, BaloteИ™ti 077015, Romania
15	Chernivtsi	Ub rid	2024-	#	26					48	26	Troian্দova St, 4, Chernivtsi, Chernivets'ka oblast, Ukraine, 58000
99	Prague	ub con	20 20	#	14.4			Opletalova 1336/35, 110 00 Praha 1-Novř© МДysto, Czechia	2023-	50	14	ЈиноДќanskřŮ 107/25, 161 00 Praha 6, Czechia
100	Kyiv	ub con	20 20	#	30.5			Holosiivskiy prospekt, 84, Kyiv, Ukraine, 02000	2023-	50	30	Vokzal'na Square, 1, Kyiv, Ukraine, 03031
101	Kyiv	ub con	20 20	#	30.3			Akademika Shalimova St, 67P', Sofiivska Borshchahivka, Kyivs'k	2023-	50	31	Holosiivskiy prospekt, 84, Kyiv, Ukraine, 02000
138	Bucharest	ub con	20 20	#	26.1				2022-	44	26	Calea GriviiИei 143, BucureИ™ti 010708, Romania
139	Bucharest	ub con	20 20	#	26.1				2022-	44	26	Strada Smřrdan 33, BucureИ™ti 030073, Romania
140	Kyiv	ub rid	2022-09-24	T18:30:00.000Z								Vokzal'na Square, 1, Kyiv, Ukraine, 03031
141	Kyiv	ub con	20 20	#	30.5				2022-	50	30	Vokzal'na Square, 1, Kyiv, Ukraine, 03031
142	Kyiv	ub con	20 20	#	30.5				2022-	50	31	Holosiivskiy prospekt, 84, Kyiv, Ukraine, 02000
185	Kyiv	ub dri	2020-	#	30.4					50	30	PICFP»PECTCŮ P\$PEPIPCbC—PSP*, 16, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
186	Kyiv	ub con	20 20	#	30.5				2020-	50	30	PICFP»PECTCŮ P\$PIIP»PePSPSCfC,CбCfPеC,PсCбP* IPiPсCбCŮ PŮiPePсCбCfCбPePсPIPс,
187	Bucharest	ub con	20 20	#	26.1				2019-	44	26	The Arch Of Triumph, PiaИia Arcul de Triumf, BucureИ™ti, P CfPJCfPSC—CŮ
188	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ P'PсPeP-P'P»CHьPSP*, 1, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
189	Kyiv	ub rid	2019-09-23	T17:12:04.000Z								PICFP»PECTCŮ P'PсPeP-P'P»CHьPSP*, 1, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
190	Kyiv	ub con	20 20	#	30.4				2019-	50	30	PICFP»PECTCŮ P\$PIIP»PePSPSCfC,CбCfPеC,PсCбP* IPiPсCбCŮ PŮiPePсCбCfCбPePсPIPс,
191	Kyiv	ub con	20 20	#	30.4				2019-	51	30	PICFP»PECTCŮ P'PЕCЕPIPCbPсPrCfCбPeP*, 69, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
192	Kyiv	ub rid	2019-	#	30.5					50	30	PICFP»PECTCŮ P'P'сCfPеP»CHьPЕC—PICFCбPeP*, 14, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
193	Kyiv	ub con	20 20	#	30.5				2019-	50	30	P'P»PеP±PсCfPЕCTCбPeP* 40 Pf, P'P»PеP±PсCfPЕCTCбPeP*, PJCfP»PECTCŮ P'P»PеP±
194	Kyiv	ub con	20 20	#	30.6				2019-	50	31	PICFP»PECTCŮ P'P'сCбPмP-PSCŮPеC—PICFCбPeP*, 30P', PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
195	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
196	Kyiv	ub con	20 20	#	30.4				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
197	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
198	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
199	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
200	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPмP»CHьPSPPеPePсPIP*, 3, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
201	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ P—PсP»PсC,PсCfCfC,C—PICFCбPeP*, 53, PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
202	Kyiv	ub con	20 20	#	30.5				2019-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
203	Kyiv	ub con	20 20	#	30.5				2018-	50	30	PICFP»PECTCŮ PьPЕC...P'PNеP»P* P'PсPJPсPSPSPCfPсPIP*, 30/1, PьPЕC—PI, PJPЕCбP°C—
204	Lviv	ub con	20 20	#	24				2018-	50	24	P'CHьPIC—PI, P'CHьPIC—PICFCбPeP* PсP±P»P'сCfC,Cб, PJPЕCбP°C—PSP*, 79000
205	Lviv	ub con	20 20	#	24				2018-	50	24	PICFP»PECTCŮ P'PIPIP»PSP* P\$PмP'PсCбPсPIP*, 11, P'CHьPIC—PI, P'CHьPIC—PICFCбPeP* P\$P
206	Kyiv	ub con	20 20	#	30.5				2018-	50	31	PьPЕC—PI, PJPЕCбP°C—PSP*, 02000
207												

Figure 4: Encoding Issues

Completeness: As Figure 5 demonstrates, there are 40 records (20%) where `begin_trip_time` and `dropoff_time` were empty, distance was 0, most fare_amount values were 0, all in line with one of the following status types: “`rider_cancelled`”, “`driver_cancelled`”, “`unfulfilled`”. Although they accurately reflect cases where the order was not completed, I decided to remove those records as well as the status field and focus solely on finished trips.

In addition, I also removed `begintrip_lat`, `begintrip_lng`, `begintrip_address`, `dropoff_lat`, `dropoff_lng`, `dropoff_address` fields as I decided not to use them in the further analysis. A snapshot of the final state of the dataset is illustrated in Figure 6. It includes 165 records and 8 fields (`city`, `product_type`, `request_time`, `begin_trip_time`, `dropoff_time`, `distance`, `fare_currency`, `fare_amount`).

	A	B	C	D	E	F	G	H	I	J
1	city	product	status	request_time	begin_trip_time	dropoff_time	distance	fare_currency	fare_amount	unit
3	Birmingham, UK	uberX	rider_canceled	2024-11-14T22:15:00.000Z			0	British Pound St	0	
5	Birmingham, UK	uberX	rider_canceled	2024-11-14T18:40:00.000Z			0	British Pound St	0	
11	Birmingham, UK	uberX	rider_canceled	2024-07-16T01:23:28.000Z			0	British Pound St	0	
12	Bucharest	uberX	rider_canceled	2024-07-15T17:29:13.000Z			0	Romanian Leu	0	
13	Bucharest	uberX	rider_canceled	2024-07-15T17:26:19.000Z			0	Romanian Leu	0	
15	Chernivtsi	UberX	rider_canceled	2024-07-07T13:54:51.000Z			0	Ukraine Hryvna	0	
28	Birmingham, UK	uberX	driver_canceled	2024-04-06T12:51:38.000Z			0	British Pound St	5.75	
39	Birmingham, UK	uberX	rider_canceled	2024-02-16T18:19:49.000Z			0	British Pound St	0	
43	Birmingham, UK	uberX	rider_canceled	2024-02-01T08:06:27.000Z			0	British Pound St	0	
54	Birmingham, UK	uberX	rider_canceled	2023-11-09T17:24:39.000Z			0	British Pound St	0	
80	Amsterdam	uberX	rider_canceled	2023-07-10T13:48:53.000Z			0	Euro	9	
81	Amsterdam	uberX	rider_canceled	2023-07-10T13:46:03.000Z			0	Euro	0	
83	Amsterdam	Comfort	rider_canceled	2023-07-09T15:19:36.000Z			0	Euro	12.65	
84	Amsterdam	uberX	rider_canceled	2023-07-09T15:16:30.000Z			0	Euro	10.45	
85	Amsterdam	uberX	unfulfilled	2023-07-09T15:11:35.000Z			0	Euro	0	
86	Amsterdam	uberX	rider_canceled	2023-07-09T15:09:16.000Z			0	Euro	19.95	
96	Birmingham, UK	uberX	unfulfilled	2023-06-17T18:43:14.000Z			0	British Pound St	0	
97	Birmingham, UK	uberX	unfulfilled	2023-06-17T18:39:54.000Z			0	British Pound St	0	
102	Brighton and Sussex	uberX	rider_canceled	2023-05-07T09:41:49.000Z			0	British Pound St	3.45	
108	Birmingham, UK	uberX	rider_canceled	2023-03-21T08:17:46.000Z			0	British Pound St	0	
117	Birmingham, UK	uberX	rider_canceled	2022-12-07T08:27:45.000Z			0	British Pound St	0	
122	Birmingham, UK	uberX	rider_canceled	2022-11-05T12:16:04.000Z			0	British Pound St	0	
123	Birmingham, UK	uberX	rider_canceled	2022-11-03T17:02:09.000Z			0	British Pound St	0	
129	Birmingham, UK	uberX	rider_canceled	2022-10-23T09:32:22.000Z			0	British Pound St	0	
136	Birmingham, UK	uberX	unfulfilled	2022-09-28T18:24:57.000Z			0	British Pound St	0	
140	Kyiv	uberX	rider_canceled	2022-09-24T18:30:00.000Z			0	Ukraine Hryvna	0	
147	Brighton and Sussex	uberX	rider_canceled	2022-07-23T11:34:53.000Z			0	British Pound St	0	
148	Brighton and Sussex	uberX	unfulfilled	2022-07-23T11:19:31.000Z			0	British Pound St	0	
149	Brighton and Sussex	uberX	rider_canceled	2022-07-23T11:01:14.000Z			0	British Pound St	0	
150	Birmingham, UK	uberX	rider_canceled	2022-07-21T21:30:11.000Z			0	British Pound St	0	
152	Birmingham, UK	uberX	rider_canceled	2022-07-10T07:34:31.000Z			0	British Pound St	0	
153	Birmingham, UK	uberX	unfulfilled	2022-07-10T07:26:55.000Z			0	British Pound St	0	
167	Birmingham, UK	uberX	rider_canceled	2021-11-14T22:22:33.000Z			0	British Pound St	0	
171	Birmingham, UK	uberX	unfulfilled	2021-08-09T10:05:45.000Z			0	British Pound St	0	
172	Birmingham, UK	uberX	rider_canceled	2021-08-09T10:03:52.000Z			0	British Pound St	0	
173	Birmingham, UK	uberX	rider_canceled	2021-08-09T09:56:14.000Z			0	British Pound St	0	
177	Birmingham, UK	uberX	rider_canceled	2021-06-12T21:24:33.000Z			0	British Pound St	0	
185	Kyiv	uberX	driver_canceled	2020-01-27T08:06:55.000Z			0	Ukraine Hryvna	40	
189	Kyiv	uberX	rider_canceled	2019-09-23T17:12:04.000Z			0	Ukraine Hryvna	0	
192	Kyiv	uberX	rider_canceled	2019-08-30T06:36:21.000Z			0	Ukraine Hryvna	0	
207										

Figure 5: Cancelled trips

1	city	product_type	request_time	begin_trip_time	dropoff_time	distance	fare_currency	fare_amount
2	Birmingham, UK	uberX	2025-01-19T19:43:08	2025-01-19T19:45:50	2025-01-19T20:14:31.000Z	8.44	British Pound Sterlir	18.92
3	Birmingham, UK	uberX	2024-11-14T21:40:57	2024-11-14T21:50:35	2024-11-14T22:14:17.000Z	15.83	British Pound Sterlir	20.95
4	Barcelona, Spain	Taxi	2024-08-29T05:43:46	2024-08-29T05:46:29	2024-08-29T06:09:25.000Z	11.62	Euro	32.05
5	Barcelona, Spain	Taxi	2024-08-27T18:36:55	2024-08-27T18:39:59	2024-08-27T18:50:29.000Z	1.5	Euro	8
6	Barcelona, Spain	Taxi	2024-08-27T14:19:08	2024-08-27T14:23:51	2024-08-27T14:40:24.000Z	3.57	Euro	14.05
7	Birmingham, UK	Comfort	2024-08-09T22:51:06	2024-08-09T22:54:48	2024-08-09T23:15:46.000Z	8.05	British Pound Sterlir	19.44
8	Birmingham, UK	uberX	2024-07-16T01:25:10	2024-07-16T01:32:03	2024-07-16T01:47:42.000Z	6.54	British Pound Sterlir	17.99
9	Bucharest, Romania	uberX	2024-07-15T08:55:06	2024-07-15T08:58:15	2024-07-15T09:25:05.000Z	12.65	Romanian Leu	71.97
10	Birmingham, UK	uberX	2024-06-19T12:35:44	2024-06-19T12:39:11	2024-06-19T12:47:16.000Z	2.26	British Pound Sterlir	6.92
11	Birmingham, UK	uberX	2024-05-09T20:59:07	2024-05-09T21:03:18	2024-05-09T21:22:32.000Z	7.6	British Pound Sterlir	13.99

Figure 6: Final State

2. Database Schema Design with SQL as a DDL

The second part of the assignment was more challenging to do as I have never created a star schema before. I created a central fact table called trip_fact and a dimension called city. Since the city field in the spreadsheet now contains both the city and country, I had to separate them in the

table. Other fields like currency or product_type are single columns, so creating dimension tables for them seemed excessive. The biggest challenge was to handle the time fields as I was uncertain whether to include them in a times dimension table or leave in the fact time. I eventually decided on the latter.

Figure 7 includes the ER diagram created in mySQL, and Figure 8 demonstrates the code to create the tables.

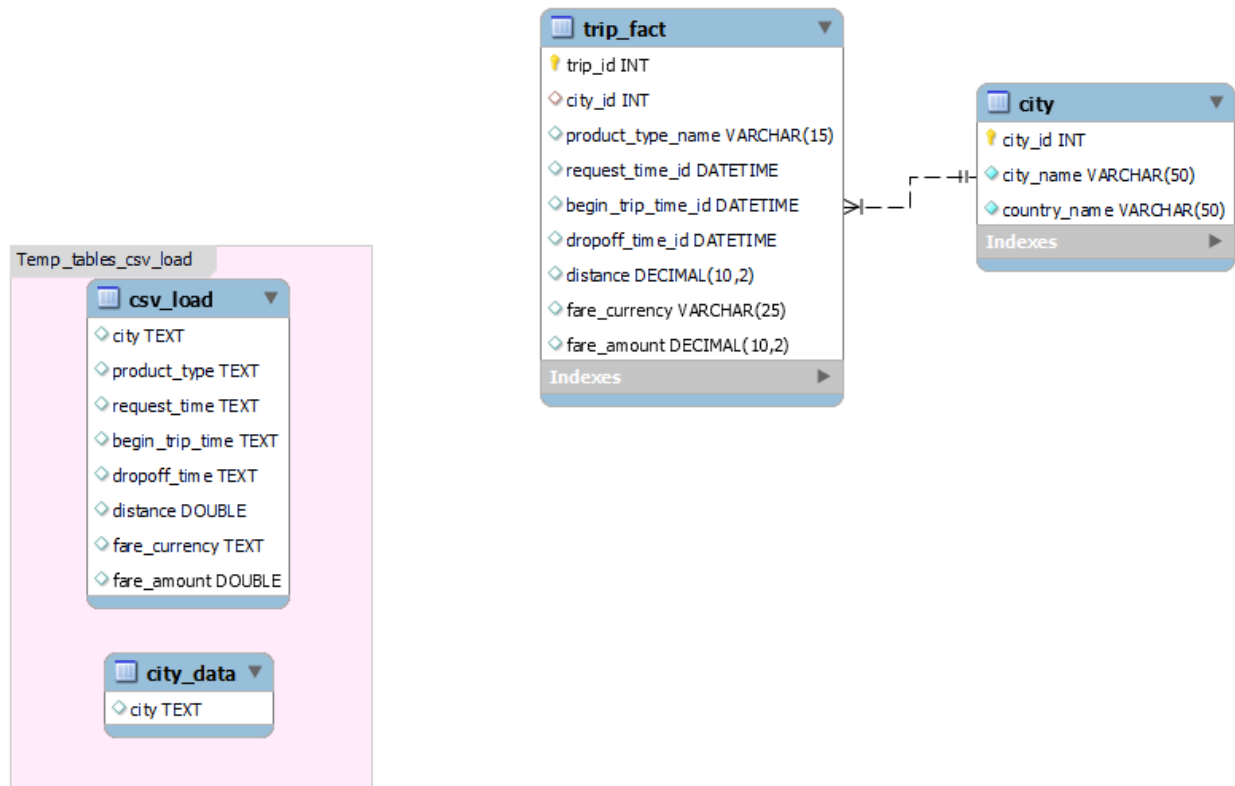


Figure 7: ER Diagram

```

1 • CREATE DATABASE Uber_trips;
2 • USE Uber_trips;
3
4 • CREATE TABLE city (
5     city_id INT AUTO_INCREMENT PRIMARY KEY,
6     city_name VARCHAR(50) NOT NULL,
7     country_name VARCHAR(50) NOT NULL
8 );
9
10 • CREATE TABLE trip_fact (
11     trip_id INT AUTO_INCREMENT PRIMARY KEY,
12     city_id INT,
13     product_type_name VARCHAR(15),
14     request_time_id DATETIME,
15     begin_trip_time_id DATETIME,
16     dropoff_time_id DATETIME,
17     distance DECIMAL(10, 2),
18     fare_currency VARCHAR(25),
19     fare_amount DECIMAL(10, 2),
20     FOREIGN KEY (city_id) REFERENCES city(city_id)
21 );
22
23 • INSERT INTO city (city_name, country_name)
24 SELECT DISTINCT
25     TRIM(SUBSTRING_INDEX(city, ',', 1)) AS city_name,
26     TRIM(SUBSTRING_INDEX(city, ',', -1)) AS country_name
27 FROM city_data;
28
29 • SELECT * FROM city;
30
31 • INSERT INTO trip_fact (
32     city_id,
33     product_type_name,
34     request_time_id,
35     begin_trip_time_id,
36     dropoff_time_id,
37     distance,
38     fare_currency,
39     fare_amount
40 )
41 SELECT
42     (SELECT city_id
43      FROM city
44      WHERE city_name = TRIM(SUBSTRING_INDEX(city, ',', 1))
45            AND country_name = TRIM(SUBSTRING_INDEX(city, ',', -1))
46     ) AS city_id,
47     product_type,
48     STR_TO_DATE(request_time, '%Y-%m-%dT%H:%i:%s.000Z') AS request_time_id,
49     STR_TO_DATE(begin_trip_time, '%Y-%m-%dT%H:%i:%s.000Z') AS begin_trip_time_id,
50     STR_TO_DATE(dropoff_time, '%Y-%m-%dT%H:%i:%s.000Z') AS dropoff_time_id,
51     distance,
52     fare_currency,
53     fare_amount
54 FROM csv_load;

```

Figure 8: SQL script

Trip_id and city_id are automatically generated unique identifiers. As Figure 9 shows, the city table contains unique city names, while the city and country names are split into two columns.

	city_id	city_name	country_name
▶	1	Birmingham	UK
	2	Barcelona	Spain
	3	Bucharest	Romania
	4	Madrid	Spain
	5	London	UK
	6	Amsterdam	Netherlands
	7	Prague	Czech Republic
	8	Kyiv	Ukraine
	9	South West	UK
	10	Dublin	Ireland
	11	Lviv	Ukraine

Figure 9: City Table

Due to the LOAD DATA INFILE limitations in mySQL, I loaded the data from the spreadsheet into temporary tables first and used SQL commands to insert the records in the appropriate tables. A snapshot of the trip_fact table is shown in Figure 10.

trip_id	city_id	product_type_name	request_time_id	begin_trip_time_id	dropoff_time_id	distance	fare_currency	fare_amount
▶ 1	1	uberX	2025-01-19 19:43:08	2025-01-19 19:45:50	2025-01-19 20:14:31	8.44	British Pound Sterling	18.92
2	1	uberX	2024-11-14 21:40:57	2024-11-14 21:50:35	2024-11-14 22:14:17	15.83	British Pound Sterling	20.95
3	2	Taxi	2024-08-29 05:43:46	2024-08-29 05:46:29	2024-08-29 06:09:25	11.62	Euro	32.05
4	2	Taxi	2024-08-27 18:36:55	2024-08-27 18:39:59	2024-08-27 18:50:29	1.50	Euro	8.00
5	2	Taxi	2024-08-27 14:19:08	2024-08-27 14:23:51	2024-08-27 14:40:24	3.57	Euro	14.05

Figure 10: Trip_fact Table